

SENTHAMIZHVELAN M

Cloud & DevOps Engineer — AWS — Docker — Terraform — CI/CD — Kubernetes

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Professional Summary

Cloud & DevOps Engineer with hands-on experience building and deploying AWS applications using Linux, Docker, GitHub Actions, Terraform, Python, monitoring, and infrastructure automation. Built an event-driven CloudOps incident-response platform with automated remediation, secure CI/CD using GitHub OIDC, AWS Systems Manager, monitoring, and failure-injection testing. B.Tech graduate in Artificial Intelligence & Data Science with practical experience in RAG, LLM APIs, MCP, and workflow automation.

Technical Skills

Cloud / AWS	EC2, VPC, IAM, S3, ECR, EBS, RDS, CloudWatch, EventBridge, API Gateway, SSM, SNS, SQS, Secrets Manager, ELB, Auto Scaling, ECS, EKS, Route 53, CloudFront, WAF
DevOps / CI/CD	Git, GitHub, GitHub Actions, Jenkins, Docker, Docker Compose, Kubernetes, Terraform, GHCR
Linux / Networking	Ubuntu Linux, Bash, SSH, Nginx, TCP/IP, DNS, HTTP/HTTPS, Subnets, Routing, Security Groups, Load Balancing
Monitoring / Security	Prometheus, Grafana, CloudWatch, CloudTrail, IAM Least Privilege, Health Checks, Secrets Management, Incident Response, Trivy
Programming	Python, FastAPI, boto3, pytest, SQL
AI / Automation	RAG, Agentic AI, LLM APIs, MCP, n8n Workflow Automation

Experience

Hema's Enterprises Private Limited (HEPL)

Feb 2026 – Aug 2026

AI Intern

- Automated internal business workflows using n8n, reducing manual processing time by **20%**.
- Built and deployed RAG-based internal knowledge retrieval systems integrated with REST APIs.
- Implemented Model Context Protocol (MCP) integrations connecting LLM workflows with internal tools and data sources.

Projects

CloudGuardian – AI-Assisted CloudOps Incident Response Platform

GitHub

AWS, Docker, GitHub Actions, Terraform, Python, FastAPI, Prometheus

- Designed an event-driven AWS incident-response workflow using **CloudWatch, EventBridge, API Gateway, FastAPI, Docker, EC2, and SSM** to process infrastructure alarms through detection, diagnosis, approval, remediation, and verification.
- Built CI/CD with **GitHub Actions**, integrating Ruff, pytest, and Trivy security scanning; successful builds publish Docker images to Amazon ECR and deploy to EC2 through AWS Systems Manager.
- Implemented **GitHub OIDC authentication** with short-lived AWS credentials, eliminating long-lived AWS access keys from the CI/CD pipeline.
- Developed guardrailed AI-assisted remediation using runbook retrieval and risk-classified, allowlisted actions executed through SSM with human approval and recovery verification.
- Added **Prometheus metrics, health/readiness endpoints, CloudWatch monitoring, 10 incident runbooks**, and Bash failure-injection scripts for CPU, memory, disk, and service failures.

ExamPrep – AI Question Paper Generator

GitHub

AWS EC2, Docker, GitHub Actions, GHCR, Nginx, React, Vite

- Containerized a React/Vite application using a **multi-stage Dockerfile**, producing an approximately **27 MB production image** served through Nginx on Ubuntu EC2.
- Built GitHub Actions CI/CD publishing Docker images to **GitHub Container Registry** using latest and commit-SHA tags for version tracking and rollback.
- Automated EC2 deployment over SSH with container restart policies and post-deployment health checks, completing deployments in approximately **one minute**.
- Integrated **Gemini and Llama 3.3 70B through Groq** with provider failover for syllabus analysis and AI-powered question generation.
- Implemented PDF and Word export functionality, managed deployment credentials through GitHub Secrets, and troubleshoot EC2 security groups, Docker permissions, and container replacement issues.

Education

C.K. College of Engineering and Technology, Cuddalore

2026

Bachelor of Technology – Artificial Intelligence & Data Science

CGPA: 8.5/10

Anandhan Memorial Matric Higher Secondary School

Higher Secondary (12th): 87.53% — Secondary (10th): 84.6%